Trials@uspto.gov
571-272-7822

Paper 37
Date: July 13, 2023

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

EYENOVIA, INC.,
Petitioner,

v.

SYDNEXIS, INC.,
Patent Owner.

IPR2022-00384
Patent 10,842,787 B2

Before JOHN G. NEW, SUSAN L. C. MITCHELL, and JAMIE T. WISZ, *Administrative Patent Judges*.

MITCHELL, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Denying Patent Owner's Motion to Exclude
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. Background and Summary

On December 29, 2021, Eyenovia, Inc. ("Petitioner") filed a Petition requesting an *inter partes* review of claims 1, 2, 4–9, and 12–19 of U.S. Patent No. 10,842,787 B2 (Ex. 1001, "the '787 patent"). Paper 1, ("Petition" or "Pet."). Sydnexis, Inc. ("Patent Owner") filed a Preliminary Response to the Petition on April 18, 2022. Paper 6 ("Prelim. Resp."). With our authorization, Petitioner filed a Reply to the Preliminary Response addressing the public accessibility of Exhibit 1004[1] and Patent Owner's proposed claim construction for "buffer" and "buffering agent." *See* Ex. 3001; Paper 7. Also, with our authorization, Patent Owner filed a Sur-reply in response. *See* Ex. 3001; Paper 8.

On July 15, 2022, we granted institution of an *inter partes* review of claims 1, 2, 4–9, and 12–19 of the '787 patent on all grounds set forth in the Petition. *See* Paper 9 ("Dec.") 2, 41.

Patent Owner filed a Response on October 21, 2022, *see* Paper 19 ("PO Resp.), and Petitioner filed a Reply on January 13, 2023, *see* Paper 21 ("Reply"). Patent Owner filed its Sur-Reply on February 24, 2023. Paper 24 ("Sur-Reply"). An oral hearing was held on April 14, 2023, and a transcript of this hearing was entered into the record. Paper 36 ("Tr.").

Patent Owner filed a Motion to Exclude Evidence on March 24, 2023. Patent Owner seeks to exclude Exhibit 1002, the Declaration of Dr. Byrn, Petitioner's declarant, Exs. 1004 and 1057 constituting the Akorn labels, and Exhibits 1079 through 1088, Eyenovia's Reply exhibits. Paper 29.

---

[1] Atropine Sulfate Ophthalmic Solution, USP 1%, NDA 206289 Product Label (July 2014) (Ex. 1004, "Akorn").

Petitioner opposed the motion, *see* Paper 30, and Patent Owner filed a reply, *see* Paper 32. For the reasons set forth herein, the motion is *denied*.

This is a Final Written Decision under 35 U.S.C. § 318(a) as to the patentability of the challenged claims on which we instituted trial. Based on the complete record before us, we determine as set forth below that Petitioner has shown by a preponderance of the evidence that claims 1, 2, 4–9, and 12–19 of the ’787 patent are unpatentable.

B. Real Parties in Interest

The parties identify themselves as the real parties-in-interest. Pet. 63; Paper 4, 2; Paper 27, 2.

C. Related Matters

Petitioner asserts "[t]here are no other judicial proceedings that may affect or be affected by a decision in this proceeding." Pet. 63. Petitioner further states that the ’787 patent issued from U.S. Application No. 15/568,381, a national phase application of International Application No. PCT/US2016/029222, which in turn is a continuation-in-part application of U.S. Application No. 14/726,139 that issued as U.S. Patent No. 9,421,199 ("the ’199 patent"). *Id.* at 63–64; *see also* Paper 11, 1 (Patent Owner provides a similar recitation of the ’787 patent's history). According to Petitioner, the ’199 patent was challenged in IPR2021-00439 by Nevakar, Inc. *Id.* at 64. Petitioner states that IPR2021-00439 and other petitions filed by Nevakar, Inc. challenging Patent Owner's patents were terminated due to settlement prior to institution. *Id.*; *see also* Paper 11, 2 (Patent Owner provides a similar recitation of the Nevakar proceedings).

Patent Owner identifies IPR2022-00414 and IPR2022-00415 as related matters. Paper 4, 2; Paper 27, 2. Patent Owner also states that it is the owner of the following United States patents related to the ’787 patent: U.S. Patent No. 10,940,145; U.S. Patent No. 10,888,557; U.S. Patent No. 9,770,447; U.S. Patent No. 10,076,515; and U.S. Patent No. 10,201,534. Paper 11, 1–2. Patent Owner also identifies the following pending U.S. Applications related to the ’787 patent: U.S. Application No. 17/097,930; U.S. Application No. 16/785,411; U.S. Application No. 16,908,426; U.S. Application No. 17,681,560; U.S. Application No. 17/721,838; and U.S. Application No. 17/721,831. *Id*. at 2.

D. The ’787 Patent

The ’787 patent relates to stabilized ophthalmic compositions comprising a muscarinic receptor antagonist for the treatment of an ophthalmic disorder, such as pre-myopia, myopia, or the progression of myopia. Ex. 1001, 5:4–8. Myopia is characterized by an axial elongation of the eye that begins during grade school years and progresses until growth of the eye is complete. *Id.* at 10:44–47. The Specification explains that muscarinic antagonists, such as atropine, prevent or arrest the development of myopia in humans. *Id.* at 9:50–56.

The Specification states that the clinical use of atropine for therapy had been limited due to side effects, such as glare from pupillary dilation and blurred vision. Ex. 1001, 12:21–24. The ’787 patent attributes these effects to high concentrations of atropine (e.g., 1 wt % or higher). *Id.* at 12:24–29. The Specification explains also that some muscarinic antagonist compositions were formulated with lower pH values (e.g., less than 4.5) to promote stability of the muscarinic antagonist. *Id.* at 10:3–7. The lower pH values, however, cause discomfort or other side effects, such as pain or a burning sensation in the eye, which can be prevented or alleviated by formulating muscarinic antagonist compositions at higher pH ranges. *Id.* at 10:7–11.

According to the Specification, "there is a need for a stabilized ophthalmic composition with extended shelf life upon storage," and there is a need for stabilizing such a composition by arresting or reducing hydrolysis of at least some of its active agents. Ex. 1001, 9:45–50. The Specification also recognizes that "there is a need for an ophthalmic composition that provides convenient and effective delivery of a muscarinic antagonist such as atropine in the eye of a patient." *Id.* at 9:50–53. In view of this, the '787 patent contemplates low concentrations of ophthalmic agents (e.g., from about 0.001 wt% to about 0.5 wt%). *Id.* at 12:30–35. For instance, the Specification describes ophthalmic compositions that include from about 0.001 wt% to about 0.05 wt% of a muscarinic antagonist, such as atropine or atropine sulfate, and water. *Id.* at 6:60–63, 7:1–2.

Additionally, the Specification describes pH values for the ophthalmic compositions (e.g., less than about 7.3, or less than about 6.4) after an extended period of time under storage condition. Ex. 1001, 7:9–16. The Specification also describes the compositions in terms of the concentration of the muscarinic antagonist based on initial concentration after an extended period of time under storage condition, and potency after an extended period of time under storage condition. *Id.* at 7:3–8, 17–21. The Specification describes "the extended period of time" as one having a duration of a range between about one week to about five years. *Id.* at 7:22–27. The storage condition is described for some embodiments as having a storage temperature of about 25° C, 40° C, or about 60° C, and in other embodiments from about 2° C to 10° C, or from about 16° C to about 26° C. *Id.* at 7:28–32.

The Specification explains that certain embodiments of the invention may further comprise additional agents, including, an osmolarity adjusting agent, a preservative, a buffer agent, a tonicity adjusting agent, a pH adjusting agent, and a pharmaceutically acceptable carrier. Ex. 1001, 7:43–8:39.

E. Illustrative Claims

Petitioner challenges claims 1, 2, 4–9, and 12–19 of the ’787 patent. Claim 1 is the only independent claim. For this Final Written Decision, independent claim 1 (set forth below with bracketed letters to identify specific limitations for ease of reference) is illustrative of the claimed subject matter.

> 1. 1[a] A stabilized ophthalmic composition for treating pre-myopia, myopia, or progression of myopia, comprising[:]
>
> 1[b] from about 0.001 wt% to about 0.05 wt% of atropine or atropine sulfate and water, wherein
>
> 1[c] the stabilized ophthalmic composition further comprises a buffering agent to provide a pH of from about 4.8 to about 6.4, wherein the
>
> 1[d] stabilized ophthalmic composition is a liquid, and wherein
>
> 1[e] the stabilized ophthalmic composition comprises less than about 10% of a degradant formed from degradation of the atropine or atropine sulfate after an extended period of time of at least 2 weeks under a storage temperature of from about 20° C. to about 70° C. and relative humidity from about 50% to about 80%.

Ex. 1001, 97:32–44 (bracketing, labels, and formatting added).

Dependent claims 2, 4–9, and 12–15 provide additional limitations to further refine the stabilized ophthalmic compositions of claim 1. *See* Ex. 1001, 97:45–48, 97:51–98:17, 98:25–41. Dependent claim 16 recites that the stabilized ophthalmic composition of claim 1 is topically administered. *See id.* at 98:42–44. Dependent claim 17 recites that the stabilized ophthalmic composition of claim 1 is administered by instillation. *See id.* at 98:45–47. Dependent claim 18 recites that the stabilized ophthalmic composition of claim 1 is administered through an eye drop bottle. *See id.* at 98:48–51. Claim 19 recites a method of treating the pre-myopia, myopia, or progression of myopia by administering an effective amount of the stabilized ophthalmic composition of claim 1. *See id.* at 98:52–56.

F. Prior Art and Asserted Grounds

Petitioner asserts that claims 1, 2, 4–9, and 12–19 would have been unpatentable on the following grounds:

Claims Challenged	35 U.S.C. §[2]	Reference(s)/Basis
1, 2, 4–9, 12–19	103	Chia,[3] Akorn,[4] Kondritzer[5]
1, 2, 4–9, 12–19	103	Chia, Akorn, Lund[6]

[2] The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112–29, 125 Stat. 284 (2011), amended 35 U.S.C. §§ 102 and 103, effective March 16, 2013. Because the application from which the ’787 patent issued has an effective filing date after that date, the AIA version of § 103 applies.

[3] Audrey Chia et al., *Atropine for the Treatment of Childhood Myopia: Safety and Efficacy of 0.5%, 0.1%, and 0.01% Doses (Atropine for the Treatment of Myopia 2)*, 119 OPHTHALMOLOGY 347−354 (2012) (Ex. 1003, “Chia”).

[4] Atropine Sulfate Ophthalmic Solution, USP 1%, NDA 206289 Product Label (July 2014) (Ex. 1004, “Akorn”).

[5] Albert A. Kondritzer & Peter Zvirblis, *Stability of Atropine in Aqueous Solution*, 46 J. AM. PHARM. ASSOC. 531−535 (1957) (Ex. 1005, “Kondritzer”).

[6] Walter Lund & Tor Waaler, *The Kinetics of Atropine and Apoatropine in AqueousSolutions*, 22 ACTA CHEM. SCAND. 3085-97 (1968) (Ex. 1007, “Lund”).

Claims Challenged	35 U.S.C. §[2]	Reference(s)/Basis
1, 2, 4–9, 12–19	103	Akorn, Wu,[7] Kondritzer

Petitioner also relies upon the Declaration of Stephen Byrn, Ph.D. (Ex. 1002). Patent Owner relies upon the Declaration of Paul A. Laskar, Ph.D. (Ex. 2003).

II. ANALYSIS

A. Principles of Law

1. Burden

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016 (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). Therefore, in an *inter partes* review, the burden of proof is on the Petitioner to show that the challenged claims are unpatentable, and that burden never shifts to the patentee. *See* 35 U.S.C. § 316(e); *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1375 (Fed. Cir. 2016) (citing *Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015)).

[7] WO 2014/182620 A1, published November 13, 2014 (Ex. 1006, "Wu").

2. Obviousness

To ultimately prevail in its challenge to Patent Owner's claims, Petitioner must demonstrate by a preponderance of the evidence[8] that the claims are unpatentable. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d). A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed subject matter and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains ("POSA" or "POSITA"). *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In determining obviousness when all elements of a claim are found in various pieces of prior art, "the factfinder must further consider the factual questions of whether a person of ordinary skill in the art would be motivated to combine those references, and whether in making that combination, a person of ordinary skill would have had a reasonable expectation of success." *Dome Patent L.P. v. Lee*, 799 F.3d 1372, 1380 (Fed. Cir. 2015); *see also WMS Gaming, Inc. v. Int'l Game Tech.*, 184 F.3d 1339, 1355 (Fed. Cir. 1999) ("When an obviousness determination relies on the combination

[8] The burden of showing something by a preponderance of the evidence requires the trier of fact to believe that the existence of a fact is more probable than its nonexistence before the trier of fact may find in favor of the party who carries the burden. *Concrete Pipe & Prods. of Cal., Inc. v. Constr. Laborers Pension Tr. for S. Cal.*, 508 U.S. 602, 622 (1993).

of two or more references, there must be some suggestion or motivation to combine the references.”). “Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant’s disclosure.” *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988); *see also In re Magnum Oil Tools*, 829 F.3d at 1381, 1381 (finding a party that petitions the Board for a determination of unpatentability based on obviousness must show that “a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.”) (internal quotations and citations omitted).

An obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

> When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. Section 103 “bars patentability unless ‘the improvement is more than the predictable use of prior art elements according to their established functions.’” *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with the above-stated principles.

B. Level of Ordinary Skill in the Art

We consider the asserted grounds of unpatentability in view of the understanding of a person of ordinary skill in the art. *KSR*, 550 U.S. at 399 (stating that obviousness is determined against the backdrop of the scope and content of the prior art, the differences between the prior art and the claims at issue, and the level of ordinary skill in the art). Factual indicators of the level of ordinary skill in the art include "the various prior art approaches employed, the types of problems encountered in the art, the rapidity with which innovations are made, the sophistication of the technology involved, and the educational background of those actively working in the field." *Jacobson Bros., Inc. v. U.S.*, 512 F.2d 1065, 1071 (Ct. Cl. 1975); *see also Orthopedic Equip. Co. v. U.S.*, 702 F.2d 1005, 1011 (Fed. Cir. 1983) (quoting with approval *Jacobson Bros.*).

Petitioner asserts that "[a] person of ordinary skill in the art ('POSA') at the time of the purported invention would have had a Ph.D. in chemistry, organic chemistry, physical chemistry, or pharmaceutics, with several years of experience preparing and/or testing pharmaceutical formulations." Pet. 21–22 (citing Ex. 1002 ¶ 62). Petitioner further contends that "[a] POSA would have been familiar with common inactive ingredients used in aqueous pharmaceutical formulations and the basic characteristics of aqueous formulations such as stability, and would have had knowledge about drug degradation kinetics." *Id.* at 22.

Patent Owner asserts that a narrower definition of a POSA should be employed—one that includes expertise in ophthalmic formulation. PO Resp.

4.[9] Patent Owner states that Petitioner's declarant, Dr. Byrn, has "extremely limited experience with ophthalmic formulation," which "calls into serious question whether he is qualified to opine on the perspective of a POSA in the relevant field." *Id*. at 5.

We do not agree with Patent Owner that the level of ordinary skill in the art should be limited to experience with ophthalmic formulation. The claims at issue generally require a liquid ophthalmic formulation of a small amount of atropine that is stabilized using buffers that can be stored for at least two weeks under certain temperature and humidity conditions. *See* Ex. 1001, 97:32–44 (claim 1). As Dr. Byrn points out, the technology surrounding use of atropine, a nonselective muscarinic antagonist, to treat myopia, has been extensively studied for over a hundred years and is known as "the oldest and most effective pharmacological treatment to inhibit the development of myopia." *See* Ex. 1002 ¶ 26 (quoting Ex. 1008, 1) (citing Ex. 1003, 1; Ex. 1012, 2). The degradation of atropine is also well known to primarily result from cleavage of the acyl-oxygen bond during ester hydrolysis, *see id*. ¶ 43 (citing Ex. 1017, 1; Ex. 1024, 5), with the rate of acid-catalyzed reaction being much slower than that of the base-catalyzed reaction, leading to stability of atropine at a lower pH range (pH < ~3), *see id*. ¶ 45 (citing Ex. 1005, 1,4–5). Concerning specifically the use of atropine in the eye, Dr. Byrn points out that it is well known that ophthalmic solutions should be "formulated to be sterile, isotonic and buffered for stability and comfort." *See id*. ¶ 47 (quoting Ex. 1019, 52) (noting Ex. 1019, 54 (pH of tear fluid is 7.4).

[9] Although Patent Owner disagrees with Petitioner's definition of a POSA, Patent Owner does not provide its own definition apart from including expertise in ophthalmic formulation. *See* PO Resp. 4–7.

None of this chemistry concerning the stability of atropine or its administration in the eye is remarkable or not well understood by a POSA as defined by Petitioner. *See* PO Resp. 10 (stating "the stability of atropine was understood to be a 'predictable function,' and a 'routine matter,' of temperature and solution pH"), 43–44 (same). We see no reason why someone with drug formulation experience, but not specific experience with ophthalmic solutions, would not qualify as a POSA. Therefore, we continue to apply Petitioner's definition of a POSA in this final written decision.

C. Claim Construction

The Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 100(b) (2019). Under that standard, claim terms "are generally given their ordinary and customary meaning" as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc) (quoting *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996)). "In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence." *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17).

Petitioner states that "[f]or the purposes of this proceeding only, Petitioner does not believe that any claim terms need to be construed." Pet. 22. Patent Owner "agrees that the claims should be understood according to their ordinary and customary meaning." PO Resp. 25–26. In reply, Petitioner asserts that the ordinary meaning of "about" is "approximately," and the ordinary meaning of "buffering agent" or "buffer" "is nothing more than an excipient that helps control pH." Reply 2. Petitioner asserts that Patent Owner agrees with the definition of "buffering agent" and "buffer" by stating that "[t]he patent uses terms buffer and buffering in their normal sense, which entails providing pH control or maintenance." *Id.* (quoting PO Resp. 29).

Based upon our review of the evidence of record, we determine that no claim terms require express construction. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (Only those terms that are in controversy need be construed, "and only to the extent necessary to resolve the controversy."). Insofar as the parties dispute whether Akorn discloses a buffer, we address those contentions in our discussion below of the challenged claims.

D. Obviousness over Chia, Akorn, and Kondritzer

Petitioner asserts that claims 1, 2, 4–9, and 12–19 would have been obvious over the combined teachings of Chia, Akorn, and Kondritzer. Pet. 23–42. In addition to challenging the public availability of Akorn, Patent Owner disagrees with the substance of Petitioner's argument and states that the Akorn reference "does not disclose the phosphate salts are buffering agents. Nor does it disclose that the Akorn product has a pH range of 3.5–6.0. Each of these failures is case-dispositive." PO Resp. 1.

1. Chia (Ex. 1003)

Chia is a journal article that describes two studies regarding the treatment of myopia using atropine eyedrops. Ex. 1003, 1. In the first study, Atropine for the Treatment of Myopia 1 ("ATOM1"), it was shown that atropine 1% eyedrops were effective in controlling myopic progression but with visual side effects resulting from cycloplegia and mydriasis. *Id.* The second study described in Chia, Atropine for the Treatment of Myopia 2 ("ATOM2"), compared the efficacy and visual side effects of 3 lower doses of atropine: 0.5%, 0.1%, and 0.01%. *Id*. The authors of Chia conclude "[a]tropine 0.01% has minimal side effects compared with atropine at 0.1% and 0.5%, and retains comparable efficacy in controlling myopia progression." *Id*.

Chia notes that atropine, a nonspecific muscarinic antagonist, at 1.0% and 0.5% has been shown to be effective in slowing myopia progression, but its safety profile regarding pupil size and accommodation was a concern. Ex. 1003, 6. Chia states:

> Every unit increase in pupil size results in an exponential increase in the amount of light entering the eye, and this can cause glare and potential phototoxicity. Atropine also decreases accommodation amplitude and near vision so that children may require bifocal or progressive glasses to read. The ideal atropine dose would be one with the best balance between efficacy and safety.

Id.

To explore the ideal atropine dose, Chia performed the ATOM2 study using 0.01% atropine as a potential control due to its assumed minimal effect. Ex. 1003, 7. Chia found that "contrary to expectations, atropine 0.01% also had significant clinical effects as evidence by its effect on myopia progression, accommodation, and pupil size." *Id*. In comparing the 0.01% atropine dose to the 0.5% and the 0.1% doses, Chia found "the difference in myopia progression at 2 years in the 0.01% group was statistically significant compared with the 0.5% group. Likewise, the difference in axial length increase was statistically larger than in both the 0.1% and 0.5% groups. However, absolute differences between groups were clinically small . . . ." *Id.* Chia did note that no other atropine 0.01% studies were available for direct comparison, but Chia did compare the data to a study[10] testing 0.05% atropine and a study[11] testing 0.025% atropine. *Id.*

Although Chia found that "[o]verall, atropine-related adverse effects were uncommon at the 0.01% dose . . . [t]here are no long-term studies on the effect of atropine on the eye, and continued vigilance is necessary. However, atropine has been clinically available since the early 1900s, and so far there are no known long-term adverse effects associated with its use." Ex. 1003, 7. Chia states:

> In conclusion, our results suggest that 0.5%, 0.1%, and 0.01% atropine remain effective in reducing myopia progression, compared with placebo treatment, and that the clinical differences in myopia progression among these 3 groups are small. The lowest concentration of 0.01% atropine thus seems to retain efficacy and is a viable concentration for reducing myopia progression in children, while attaining a clinically significant improved safety profile in terms of accommodation, pupil size, and near visual acuity, and subsequently reduced adverse impact on visual function. Moreover, the 0.01% formulation exhibited fewer adverse events. Atropine 0.01% is currently not commercially available. However, these findings collectively suggest that a nightly dose of atropine at 0.01% seems to be a safe and effective regimen for slowing myopia progression in children, with minimal impact on visual function in children.

Id. at 7–8.

---

[10] Jong-Jer Lee et al., *Prevention of myopia progression with 0.05% atropine solution*," 22 J. OCUL. PHARMACOL. & THER. 41–46 (2006) (Ex. 1012, "Lee").

[11] Fang et al., *Prevention of myopia onset with 0.025% atropine in premyopic children*, 26 J. OCUL PHARMACOL. & THER. 341–45 (2010) (Ex. 1013, "Fang").

2. Akorn (Ex. 1004)

Petitioner identifies Ex. 1004 as a product label for Akorn Atropine Care, which appears to be an excerpt from New Drug Application 206289. *See* Pet. vii; Ex. 1004, 1. Akorn describes a 1% atropine sulfate ophthalmic solution for topical application to the eye. Ex. 1004, 1. Akorn explains that atropine is an anti-muscarinic agent that is indicated for cycloplegia, mydriasis, and penalization of the healthy eye in the treatment of amblyopia. *Id.* at 1, 2. Akorn explains atropine's mechanism of action as follows:

> The pupillary constrictor muscle depends on muscarinic cholinoceptor activation. This activation is blocked by topical atropine resulting in unopposed sympathetic dilator activity and mydriasis. Atropine also weakens the contraction of the ciliary muscle, or cycloplegia. Cycloplegia results in loss of the ability to accommodate such that the eye cannot focus for near vision.

Id. at 4. Akorn describes applying 1 drop of the solution to the cul-de-sac of the conjunctiva forty minutes prior to an intended maximum dilation time. *Id.*

Akorn explains that the solution contains atropine sulfate as an active ingredient. Ex. 1004, 3. Akorn further describes the following inactive ingredients: "benzalkonium chloride 0.1 mg (0.01%), dibasic sodium phosphate, edetate disodium, hypromellose (2910), monobasic sodium phosphate, hydrochloric acid and/or sodium hydroxide may be added to adjust pH (3.5 to 6.0), and water for injection USP." *Id.*

3. Kondritzer (Ex. 1005)

Kondritzer is a journal article that relates to a kinetic study of the proton catalyzed hydrolysis of atropine "to identify and evaluate the factors involved in the deterioration of aqueous solutions of atropine and its salts," and confirmed the temperature dependency of the reaction. Ex. 1005, 1.

Kondritzer explains that the alkaline hydrolysis of atropine in aqueous solution involves two reactions: "one involving the unprotonated (base) form, and the other the protonated (ionic) form of the alkaloid," with each reaction being governed by hydroxyl-concentration and temperature. *Id.* Kondritzer also notes some previous work concerning the stability of aqueous atropine solutions stating: "The preservation of aqueous atropine solutions by buffering at pH 4-5 has been recommended." *Id.* at 2.

Kondritzer's evaluation included studying the hydrolysis of atropine in various perchloric acid solutions at temperatures varying from 70° C to 90° C. Ex. 1005, 2. Based on this, Kondritzer prepared an equation to predict the rate of proton hydrolysis at any particular temperature and hydrogen ion concentration. *Id.* Kondritzer takes into account the effect of both hydroxyl and hydrogen ions to determine the pH of minimum hydrolysis at various temperatures and calculates the hydrolytic rates at these minimum pH values. *Id.* Kondritzer also calculates the half lives of atropine solutions for various pH values and temperatures. *Id.*

From measuring the rate of acid hydrolysis of atropine in aqueous solution, Kondritzer concludes that the predominant catalytic reaction above pH 4.5 involves the hydroxyl ion, and the predominant catalytic reaction below pH 3 involves the hydrogen ion. Ex. 1005, 4. Kondritzer also states:

> The half lives of solutions of atropine at various pH values and temperatures can be calculated for both hydroxyl-ion and hydrogen-ion catalysis. The former accounts for the instability above approximately pH 5 and the latter for instability below approximately pH 3. The half life at any hydrogen ion concentration between pH 3 and 5 is so great that its experimental determination is not practical.

Id.

Kondritzer states that "[a]s expected, the hydrogen ion catalyzed hydrolysis of atropine is slow." Ex. 1005, 4. Based on the study, Kondritzer concludes that "[t]he pH for maximum stability of atropine in sterile aqueous solution varies between 4.11 at 0° and 3.24 at 100°." *Id.* at 5.

4. Analysis

For its assertion that claims 1, 2, 4–9, and 12–19 would have been obvious over the combined teachings of Chia, Akorn, and Kondritzer, Petitioner relies on Akorn as teaching or suggesting adding to an atropine sulfate ophthalmic liquid composition a buffering agent to "provide a pH of from about 4.8 to about 6.4," as required by claim 1. Pet. 28–31. Patent Owner challenges Petitioner's reliance on Akorn asserting that the Petition fails to prove that Akorn was publicly available prior to the undisputed critical date of April 23, 2015, which Patent Owner also raised in its Preliminary Response. *See* PO Resp. 58–61; Prelim. Resp. 29–32. Patent Owner also challenges the testimony of Dr. Byrn in its Motion to Exclude Evidence asserting that "Dr. Byrn's proffered opinions exceed his expertise, are not based on 'scientific knowledge,' 'sufficient facts or data,' and are not 'the product of reliable principles' and methods 'reliably applied' to the facts of this case." Paper 29, 2 ("Mot. to Exclude").

Because the prior art status of Akorn and Dr. Byrn's testimony is relevant to all grounds asserted by Petitioner, we begin our discussion with these issues.

a) Prior Art Status of Akorn (Ex. 1004)

Petitioner has the burden to prove Akorn qualifies as prior art. *See In re Magnum Oil Tools*, 829 F.3d at 1376. "[A]t the institution stage, the petition must identify, with particularity, evidence sufficient to establish a reasonable likelihood that the reference was publicly accessible before the critical date of the challenged patent and therefore that there is a reasonable likelihood that it qualifies as a printed publication." *Hulu, LLC v. Sound View Innovations, LLC*, IPR2018-01039, Paper 29 ("*Hulu*") at 13 (PTAB Dec. 20, 2019) (precedential).

"[P]ublic accessibility" is considered to be "the touchstone in determining whether a reference constitutes a 'printed publication' bar under 35 U.S.C. §102(b)." *In re Hall*, 781 F.2d 897, 899 (Fed. Cir. 1986). "A given reference is 'publicly accessible' upon a satisfactory showing that such document has been disseminated or otherwise made available to the extent that persons interested and ordinarily skilled in the subject matter or art exercising reasonable diligence, can locate it." *SRI Int'l, Inc. v. Internet Sec. Sys., Inc.*, 511 F.3d 1186, 1194 (Fed. Cir. 2008) (quoting *Bruckelmyer v. Ground Heaters, Inc.*, 445 F.3d 1374, 1378 (Fed. Cir. 2006)).

A determination whether a particular reference qualifies as a printed publication "is a legal determination based on underlying fact issues, and therefore must be approached on a case-by-case basis." *Hall*, 781 F.2d at 899. In a proceeding before the Board, there is no presumption in favor of finding that a reference is a printed publication. *Hulu*, Paper 29 at 16.

During the institution phase of this proceeding, we authorized further briefing on the issue of whether Akorn was publicly accessible before the critical date, and each party filed a brief addressing this issue. *See* Ex. 3001; Paper 7; Paper 8. In our Institution Decision, we addressed the parties' arguments and determined on the record before us at that time that "Petitioner has adequately shown for institution that Akorn was sufficiently accessible to the public interested in the art and searchable on the DailyMed website such that a POSA could have found it with reasonable diligence."

Dec. 27. In making this determination, we examined Petitioner's additional evidence, Exhibit 1057, that it submitted with its reply to the preliminary response, and determined "that Petitioner properly relies on Exhibit 1057 as additional evidence to confirm that Exhibit 1004 was published and publicly accessible in July of 2014." *See* Dec. 26–27.

In making this determination, we made the following findings concerning Exhibit 1057 for purposes of institution.

> In particular, on the current record, we find that Petitioner's Exhibit 1057 provides persuasive corroboration for its contention that Akorn was publicly accessible in July of 2014. Exhibit 1057 includes printouts of webpages from "DailyMed." Ex. 1057. In the "ABOUT DAILYMED" section on the first page of the exhibit, there is a statement explaining that "The National Library of Medicine (NLM)'s DailyMed searchable database provides the most recent labeling submitted to the Food and Drug Administration (FDA) by companies and currently in use (i.e., 'in use' labeling)." *Id.* at 1. Among other items, the DailyMed "contains labeling for prescription and nonprescription drugs for human and animal use." *Id.* The statement also explains that "[t]he NLM provides DailyMed to the public" and "[t]he labeling on DailyMed is typically reformatted to make them easier to read." *Id.*
>
> On the third page of Exhibit 1057, there is a copy of the DailyMed "LABELING ARCHIVES" webpage which depicts a "Labeling Archives Search," with a first field for entering a drug name, and a second field for entering a date. *Id.* at 3. The webpage explains that "[t]his archive allows the user to retrieve the label current for a given date." *Id.* Search results for the term "ATROPINE SULFATE" on "07/31/2014" are displayed. *Id.* There is an indication that the search provided 135 results which are listed in a row, apparently over a number of webpages. *Id.* The results are organized with the "DATE POSTED" identified in one column, along with the term "download" which appears to be formatted as a web link, and the "DRUG NAME" identified in a second column, wherein the drug name listed appears to be formatted as a web link. *Id.* Below the drug name, the name of

the "Packager" is identified, along with a "Version" number. *Id.* Petitioner draws our attention to the label identified with July 24, 2014, as the "DATE POSTED," "Atropine (atropine sulfate) solution/drops," as the "DRUG NAME," Akorn, Inc. as the "Packager," with the "Version" noted as "7." *Id.*

We understand the next page of the exhibit to be a copy of the DailyMed webpage depicting the version 7 label for Akorn's atropine sulfate that was posted on July 24, 2014. Ex. 1057, 4. Based on our review, but for minor web-related formatting differences, this label posted on DailyMed appears to be the same as the label depicted in Exhibit 1004. In particular, we observe that both labels: (a) refer to "HIGHLIGHTS OF PRESCRIBING INFORMATION" for "Atropine Sulfate Ophthalmic Solution, USP 1% for topical application to the eye;" (b) include the "Revised: 7/2014" statement; (c) include the same indications and usage, dosage and administration; (d) set forth the same sections/subsections; and (e) describe the contents of the drug in the same manner, including the statement identifying inactives as: "benzalkonium chloride 0.1 mg (0.01%), dibasic sodium phosphate, edetate disodium, hypromellose (2910), monobasic sodium phosphate, hydrochloric acid and/or sodium hydroxide may be added to adjust pH (3.5 to 6.0), and water for injection USP." *Compare* Ex. 1004, 1–5, *with* Ex. 1057, 4–6.

Taken together, we find that the evidence relied on in the Petition, i.e., Exhibits 1004, 1020, 1021, along with Exhibits 1056, 1058, and especially, 1057, submitted with the Reply, provide strong indicia that the July 2014 version of Akorn's atropine sulfate drug label represented on the current record as Exhibit 1004 was posted on July 24, 2014, on the DailyMed website. We do not find any dispute on the current record that DailyMed is a federal government website that is accessible to the public and presented in a manner that allows the public to readily search its database for drug labels that have been approved by the FDA. Moreover, as Petitioner asserts, the DailyMed website explains, "The DailyMed RSS feed provides updates and information about new drug labels approved by the FDA and published on NLM's DailyMed Web site." Ex. 1057, 1. In other words, the DailyMed website further extends its reach to the public through such feed.

Dec. 23–25.

Patent Owner in its Response continues to question the public availability of Akorn. *See* PO Resp. 58–60. Patent Owner continues to assail the public availability of Akorn as shown by the exhibits originally presented with the Petition, *see id.*, but we agreed with some of these criticisms. For instance, we stated:

> To begin, we note that, in some respects, we agree with Patent Owner's position. For example, we agree with Patent Owner that the "7/14" revision date printed on Akorn, on its own, indicates only that some unknown author(s) modified the document on that date. *See* Prelim. Resp.30, PO Sur-reply 2–3. Indeed, similar to the copyright or creation dates in *Hulu*, the revision date, by itself, is insufficient to demonstrate that the exhibit was publicly available on that date. *See Hulu*, Paper 29 at 19. We also agree with Patent Owner that neither the FDA approval date of the Akorn atropine sulfate drug product nor the sale or use of that product demonstrates the public accessibility of drug label reflected in Akorn prior to the critical date. *See* Prelim. Resp. 30, PO Sur-reply 2–3.

Dec. 23. Our decision on the public availability of Akorn turned on the additional evidence Petitioner submitted as Exhibit 1057. *See id.* at 23–27.

The only criticism Patent Owner presents with reference to our analysis of Exhibit 1057 as set forth above is to question a statement on the DailyMed webpage concerning when filtering by published date was added to Web Services. *See* PO Resp. 60; Sur-Reply 25. Patent Owner states: "But Petitioner's DailyMed exhibit (created on September 15, 2021) acknowledges that "Filtering by Published Date [Was] Added to Web Services" only as late as October 2015 (EX1057, 1, 2), which falls after the critical date. EX1001, Cover. As the very same search means that Petitioner used to find the label in EX1057 admittedly did not exist before the critical date, Petitioner has failed to demonstrate pre-critical date public availability of this new label." PO Resp. 60.

We find that Patent Owner's criticism misses the mark. Petitioner is using the DailyMed exhibit as further evidence that Akorn was publicly available as of the revision date of 7/2014 as shown on Akorn, Ex. 1004. *See* Reply 1 ("As set forth in the Petition, EX1004, a drug label to Akorn, bears July 2014 revision dates that comport with its FDA approval and publication well before the critical date. *See* Pet., 9; *Hulu*, 13. It also bears Akorn's corporate insignia. EX1004, 5. Additional evidence confirms these dates are reliable. EX1056-58."); 3 ("Public records from DailyMed, part of the NIH's National Library of Medicine, show that EX1004 was posted publicly on **July 24, 2014**."). Whether a POSA could have searched this way for labels published before the critical date is irrelevant; Petitioner is only using the evidence to show that Akorn was, in fact, published in July 2014, before the critical date on a searchable website. *See id*. Patent Owner does not dispute that DailyMed was searchable by drug name, which would be the logical way a POSA would search such a database for information for a drug of interest.

Petitioner also provides further evidence that the Akorn label was published on July 24, 2014, through a response by the National Library of Medicine ("NLM") to a Freedom of Information Act request. *See* Ex. 1071. In this Freedom of Information Act request, Petitioner asked for written confirmation that the substance of the Akorn label was posted on July 24, 2014 on the DailyMed website. *Id*. The National Institutes of Health ("NIH") responded that version 7 of the Akorn label was published on DailyMed on July 24, 2014, and provided the URL that is reflected in the Akorn label in Exhibit 1057 from the DailyMed website. *See* Ex. 1072, 6;

Ex. 1057, 4–7. NIH also confirmed that NLM "received version 7 of this label from the FDA on 07/24/2014 @ 2:15:04 PM," "published version 7 on DailyMed on the same day, 07/24/2014 @ 14:59:48 PM," and confirmed that "NLM does not make changes to the content contained within any SPL [or structured product labeling] records that it receives from the FDA." Ex. 1072, 6. We find that Petitioner has provided more than sufficient evidence that the Akorn label reflected in Ex. 1004 was publicly available before the critical date.

For the reasons set forth above, we determine that Petitioner has shown by a preponderance of the evidence that Akorn, Ex. 1004, is a printed publication that was publicly available before the critical date.

b) Patent Owner's Motion to Exclude Evidence

Patent Owner moves to exclude from evidence the Declaration of Dr. Byrn (Ex. 1002), Exhibits 1004 and 1057 that constitute the Akorn labels, and Exhibits 1079 through 1088 that Petitioner submitted with its Reply. Mot. to Exclude 2. Petitioner filed an Opposition (Paper 30, "Opp. to Mot. to Exclude"), and Patent Owner filed a Reply (Paper 32, "Reply to Mot. to Exclude"). For the following reasons, Patent Owner's Motion to Exclude is *denied*.

Pursuant to our Rules, a motion to exclude evidence must be filed to preserve any previously-made objections to evidence. 37 C.F.R. § 42.64(c). The motion must identify where in the record the objections were made, and must explain the objections. *Id.* Patent Owner appropriately identifies where in the record it previously served objections to the exhibits it seeks to exclude. Mot. 2 (citing Papers 12, 22).

As the moving party, Patent Owner bears the burden of proof to establish that it is entitled to the requested relief, i.e., the exclusion of evidence as inadmissible under the Federal Rules of Evidence ("FRE"). *See* 37 C.F.R. §§ 42.20(c), 42.64(a).

(1) Dr. Byrn's Testimony (Ex. 1002)

Patent Owner challenges Dr. Byrn's testimony under FRE 702 as not based on sufficient "scientific knowledge," not based on "sufficient facts or data," and not "the product of reliable principles" and methods "reliably applied" to the facts of this proceeding. Mot. to Exclude 2.

(a) Requisite Scientific Knowledge

For instance, although Dr. Byrn agrees that the '787 patent is "directed to the field of ophthalmic compositions and ophthalmic solutions," Patent Owner asserts that Dr. Byrn's curriculum vitae (Ex. 1002, Appendix A) "provides no indication that Dr. Byrn has any expertise in ophthalmic formulation." Mot. to Exclude 2–3 (citing Ex. 2009, 97:19–98:14, 104:20–106:5). Patent Owner further states:

> During his deposition, Dr. Byrn identified brief eyedrop forays (none significant enough to merit mention in his CV or declaration), but could not identify any ophthalmic product he formulated before the critical date. Even as of 2021, Dr. Byrn never had primary responsibility for formulating any drug product that has received regulatory approval or been commercialized.

Mot. to Exclude 3 (citing Ex. 2010 ¶ 8). Patent Owner also criticized Dr. Byrn's alleged lack of knowledge of ophthalmic terms such as lacrimation. *See id.*

Petitioner responds that the primary focus in evaluating the admissibility of testimony under FRE 702 is "whether an expert's opinion will assist the trier of fact," a low bar, and many of Patent Owner's criticisms go to the weight to be accorded Dr. Byrn's testimony rather than its admissibility. Opp. to Mot. to Exclude 1–6. In applying FRE 702, Petitioner asserts that "Dr. Byrn, a professor of medicinal chemistry who specializes in drug formulation, is eminently qualified to opine about the straightforward formulation issues presented in this proceeding." *Id*. at 1.

Petitioner asserts that Patent Owner does not contest that Dr. Byrn meets Petitioner's POSA definition (as we have determined should be applied in this proceeding), but even if we required experience in formulating ophthalmic pharmaceuticals, we find that Dr. Byrn has such experience. *See* Opp. to Mot. to Exclude 2–4; Ex. 1002, Appendix A ¶¶ 3–10. Petitioner asserts that Dr. Byrn testifies that:

> [H]e worked as a consultant for Alcon—a well-known eye care company—on a number of ophthalmic products, including ones that were FDA-approved, prior to the critical date. *See* Ex. 2009, 136:12–137:12. He further testified that he worked with eye drops as part of Purdue's Africa program. *Id*. at 49:149–50:2.

Id. at 4. Petitioner also asserts that Dr. Byrn accurately described the term "lacrimation" in his deposition. *Id*. at 4 n.2 (citing Ex. 2009, 49:11–14; Ex. 1002 ¶ 77 (quoting Ex. 1047, 2)).

As we have previously determined, a POSA need not have specific experience in ophthalmic formulation. *See* Section II.B. Dr. Byrn has ample experience to qualify as a POSA, which we defined as a person having a Ph.D. in chemistry, organic chemistry, physical chemistry, or pharmaceutics, with several years of experience preparing and/or testing pharmaceutical formulations at the time of the purported invention. *Id*.; *see* Ex. 1002 ¶¶ 3–10. Dr. Byrn has the requisite academic background, as well as extensive experience in drug development and formulation. *See* Ex. 1002

¶¶ 3–10. Dr. Byrn testifies that his "laboratory has done extensive research on drug formulations, and [he has] authored over two hundred book chapters and peer-reviewed journal articles on properties of a wide variety of pharmaceutical compositions." *Id.* ¶ 10.

We also find, however, that Dr. Byrn has experience with ophthalmic formulations. Dr. Byrn testifies that he has "worked quite a bit with eyedrops" in Purdue's Africa program. Ex. 2009, 49:21–24. In his Declaration, Dr. Byrn testifies that he was a co-founder of the Sustainable Medicines in Africa in the Kilimanjaro and Arusha regions of Tanzania in 2007. Ex. 1002 ¶ 5. Dr. Byrn describes this program more extensively in his curriculum vitae as providing educational programs that are "aimed at providing source of well-trained manufacturing scientists for pharmaceutical industry in Tanzania and Africa," and including a GMP-level pharmaceutical manufacturing facility to teach manufacturing under strict quality control, as well as a quality medicines laboratory equipped with HPLCs. Ex. 1002, Appendix, 101. Dr. Byrn also testifies that he consulted with Alcon on a number of ophthalmic drug products although he was not primarily responsible for their formulation. *See* Ex. 2009, 136:12–20.

We find Dr. Byrn's alleged lack of "primary" responsibility for the formulation of an ophthalmic product to be somewhat of a red herring in assessing whether Dr. Byrn has "expertise in ophthalmic formulation" as Patent Owner asserts should be required. *See* Mot. to Exclude 2–3. In response to questions about whether Dr. Byrn ever had primary responsibility for formulating a drug product that has received regulatory approval, Dr. Byrn testifies that he has "been on a team that formulated drug products for both for commercial sale and IND testing." Ex. 2009, 134:19–24. Dr. Byrn also pointed out what he considered to be a false premise in the questions related to "primary" responsibility for drug formulation.

Dr. Byrn testifies:

> I have been a consultant on a large number of products that came on the market and I worked in a team, and my work was – part of the work was critical for certain specifications and certain marketing steps, but I don't believe that in this day and age a person is primarily responsible for bringing a drug on the market; it's a team activity.

Id. at 135:15–23. We credit Dr. Byrn's testimony here.[12] We also determine that Dr. Byrn satisfies the requirements for a POSA as we have determined it should be defined, without specific expertise in ophthalmic formulation, but also determine that even if we applied Patent Owner's heightened standard requiring such expertise, Dr. Byrn satisfies that definition also. Dr. Byrn has the requisite scientific knowledge to serve as an expert for Petitioner.

(b) Sufficiency of Underlying Facts and Data; Reliability of Principles and Methods as Applied to Underlying Facts and Data

Patent Owner asserts that Dr. Byrn's alleged inexperience with ophthalmic formulations led him to offer an opinion not based on sufficient facts or data or reliable principles, i.e., that Dr. Byrn's opinion that "Akorn teaches the uses of buffering agents to achieve a solution pH range of 3.5 to 6.0 were 'premised' on an undisclosed assumption that the buffering range of the weak acid extends ±2 pH units away from its pK_a." Mot. to Exclude 4

[12] We also find that Dr. Byrn was clearly familiar with the term "lacrimation," which means "the secretion of tears especially when abnormal or excessive," but chose to use the word "tearing" instead. *See* Ex. 2009, 49:3–14; *see also* https://www.merriam-webster.com/dictionary/lacrimation (accessed June 25, 2023) (defining lacrimation).

(citing PO Resp. 39–40; Ex. 2009, 96:5–97:4; Ex. 1002 ¶¶ 20, 24, 65–66, 80, 87, 94–95, 99, 154–55, 158, 167, 171, 200, 201, 208, 214, 220). Patent Owner also asserts Dr. Byrn's assumption has been shown to be incorrect in a district court proceeding in which he testified. *See* Mot. to Exclude 4–7.

Petitioner counters that Dr. Byrn's opinions are based on sufficient facts and data and are reliable. Opp. to Mot. to Exclude 5–6. As an example, Petitioner points to Dr. Byrn's reliance on Remington (Ex. 1019), which Dr. Laskar admits is a formulator's "bible," to establish that "Remington means exactly what it says: buffer capacity represents a compromise between stability and comfort, and thus a buffer with diminished capacity (*e.g.,* the one provided by sodium phosphate salts) should be used." Opp. to Mot. to Exclude 5 (citing Ex. 2003 ¶¶ 62–63; Ex. 1002 ¶¶ 97–98).

Petitioner also points to support for Dr. Byrn's opinion that sodium phosphate salts "can still buffer outside of ± 1 unit away from their pK_a" in the "reliable scientific literature that Patent Owner seeks to preclude the Board from considering." Opp. to Mot. to Exclude 6 (citing Exs. 1079–1083 (buffer recipes using the claimed sodium phosphate salts below pH 6.0)). Finally, Petitioner asserts that Dr. Byrn's testimony in the district court proceeding is inapposite because

> The primary legal issue in that case dealt with infringement under the doctrine of equivalents, and whether a generic company's "proposed product contains an equivalent to the 'buffer' of the claims[.]" *See* Ex. 2010, 2. Unlike here, the alternative excipient that the generic company contended was a buffer had not be identified, disclosed, and claimed as such by the patentee. *See* Ex. 1001 (claim 7).

Id. at 6–7.

We agree with Petitioner here that Patent Owner's concerns about the sufficiency of the support for Dr. Byrn's opinions goes more to the weight to be accorded such opinions rather than a need to exclude such opinions. *See* TPG 40–41, 79 ("A motion to exclude must explain why the evidence is not admissible (e.g., relevance or hearsay) but may not be used to challenge the sufficiency of the evidence to prove a particular fact."). We also note that the findings in the district court litigation are based on a different record that is not before us here. For instance, the district court relies on a Lewis reference and a Harris reference that is not before us and testimony from Dr. Richard Moreton that is also not before us here. *See* Ex. 2010 ¶¶ 63–69. Therefore, we determine that the district court's findings are not sufficiently probative of the questions before us here.

Also, the district court's findings are not as definitive as Patent Owner states. The patent owner in the district court case attempted to prove infringement by the doctrine of equivalents for a formulation that did not expressly contain a buffer, but that it argued inherently contained a buffer from the chemical disassociation of the active ingredient in the product. *See* Ex. 2010, 64–65. The district court found that patent owner failed to prove that this actually occurs in the accused product. *Id.* at 65. The district court case is in stark contrast with the underlying facts of this proceeding in which the prior art formulation expressly contains the same buffers as claimed in the '787 patent. *See* Ex. 1001, 98:7–10 (claim 7); Ex. 1004, 3.

(c) Conclusion

Based on the foregoing, we *deny* Patent Owner's motion to exclude Exhibit 1002, Dr. Byrn's testimony.

(2) The Akorn Labels (Exs. 1004 and 1057)

Patent Owner asserts that Exhibits 1004 and 1057 should be excluded because Petitioner failed to show that Exhibit 1004 (Akorn) was publicly available before the critical date, and Exhibit 1057 is deficient to show such public availability of Exhibit 1004. Mot. to Exclude 7–8.

We have previously addressed these issues in our determination of the prior art status of Akorn, Ex. 1004, set forth above. *See* Section II.D.4.a). For these same reasons, we *deny* Patent Owner's motion to exclude Exhibits 1004 and 1057.

(3) Exhibits 1079–1088

Patent Owner moves to exclude Exhibits 1079 through 1088 submitted with Petitioner's Reply. *See* Mot. to Exclude 2, 8–15. Patent Owner asserts that these exhibits should have been submitted with the Petition and are untimely. *Id*. at 8. Patent Owner also asserts that Dr. Byrn did not "elect to submit a Reply declaration, rely upon any of these documents, or testify that any of them are the kinds of documents an expert in his field (much less a POSA in the field of the invention) would rely upon." *Id*.

Petitioner asserts that these documents were presented to Dr. Laskar at his deposition to rebut Patent Owner's assertions that "two common buffering agents, sodium dihydrogen phosphate and disodium hydrogen phosphate—identified and claimed by the '787 patent as buffering agents—are not buffers in the claimed range of about 4.8 to about 6.4." Opp. to Mot. to Exclude 10–11 (citing PO Resp. 35–36; Ex. 2003 ¶ 56).

Our Trial Practice Guide states that:

> Petitioner may not submit new evidence or argument in reply that it could have presented earlier, e.g. to make out a prima

> facie case of unpatentability. A party may also submit rebuttal evidence in support of its reply.

Patent Trial and Appeal Board Consolidated Trial Practice Guide 73 (Nov. 2019) (citing *Belden Inc. v. Berk-Tek LLC*, 805 F.3d 1064, 1077–78 (Fed. Cir. 2015).

Here, we find that Exhibits 1079 through 1088 are appropriate rebuttal evidence. Petitioner uses these exhibits to test Dr. Laskar's opinions concerning the buffering range for sodium dihydrogen phosphate and disodium hydrogen phosphate. *See generally* Ex. 1078, 91–189.[13] As the Federal Circuit noted in *Belden*, there is no bright-line demarcation between support for Petitioner's prima facie case and rebuttal evidence. For instance, the Federal Circuit stated that using rebuttal evidence as support for the prima facie case does not necessarily mean that is was "necessary" for the prima facie case requiring it to be in the Petition.

> Evidence admitted in rebuttal to respond to the patent owner's criticisms will commonly confirm the prima facie case. That does not make it necessary to the prima facie case. And nothing required the Board to write its opinion to separate the material offered by [petitioner] at different stages of the proceeding.

See Belden, 805 F.3d at 1079.

For these reasons, we *deny* Patent Owner's motion to exclude Exhibits 1079 through 1088.

---

[13] In response to Patent Owner's Objections to Evidence, Petitioner also provided a Declaration of Rebecca L. Baker, a Research Analyst at the law firm representing Petitioner, attesting to the authenticity of Exhibits 1079, 1081–1085. *See* Ex. 1095.

c) Claim 1

Petitioner asserts that claim 1 recites elements previously known in Chia and Akorn: "Chia's low-concentration atropine formulation, combined with Akorn's buffering agents to provide an aqueous atropine solution with a pH range that was well known to be desirable for treating myopia." Pet. 23. The claimed "stabilized" atropine solution of claim 1, Petitioner asserts, is an expected result of the combination of Chia's low-concentration atropine formulation and Akorn's buffering agents. *Id.*

Petitioner describes what it terms the "ample" motivation to make such a combination and the reasonable expectation of success for such a combination as follows. *See* Pet. 23 (citing Ex. 1002 ¶¶ 64–78).

> In 2012, *Chia* disclosed that low-concentration atropine eye drops effectively treated myopia and reduced patient compliance problems associated with higher concentration compositions. Ex. 1003, 7–8. *Chia*'s clinical success would have provided motivation to make and use low-concentration atropine formulations at optimal pH levels for patient comfort and stability. To determine the suitable pH levels, a POSA would have looked to FDA-approved atropine ophthalmic solutions. Ex. 1002 ¶¶ 65, 87. As a result, a POSA would have been motivated to use *Akorn*'s pH range of 3.5–6.0 for the low-concentration formulation disclosed in *Chia*. A POSA would have confirmed that this range was appropriate via routine experimentation and based on the predicted stability of atropine at various pHs disclosed in *Kondritzer*.

Pet. 23–24.

Patent Owner responds that Akorn "does not disclose the phosphate salts are buffering agents. Nor does it disclose that the Akorn product has a pH range of 3.5–6.0. Each of these failures is case-dispositive," as all grounds rely exclusively on Akorn "to disclose a stabilized atropine ophthalmic aqueous solution comprising a buffering agent to provide a pH within the claimed range." PO Resp. 1, 26. Patent Owner also asserts that Petitioner's only motivation for making the claimed combinations in the '787 patent results from improper hindsight. *See id.* at 43–52.

(1) Limitations 1[a], 1[b], 1[d], and 1[e]

We begin our analysis with the limitations of claim 1 that Patent Owner does not dispute are taught by the prior art. Petitioner points to both Chia and Akorn as teaching limitation 1[a]: "[a] stabilized ophthalmic composition for treating pre-myopia, myopia or progression of myopia." Pet. 28–29; *see* Ex. 1002 ¶¶ 83–87. Chia describes the ATOM2 study, which "examined the effect of lower doses of atropine to determine whether these concentrations could result in efficacy in preventing myopia progression, with less visual side effects (i.e., pupil dilation, loss of accommodation, and near vision blur)." Ex. 1003, 1–2. The treatment phase of the ATOM2 study lasted 24 months, *id.* at 2, and "[t]rial medications were prepackaged so that bottles were prelabeled with subject number and of similar appearance. Trial medication consisted of the appropriate dose of atropine sulfate with 0.02% of 50% benzalkonium chloride as a preservative . . . ." *Id.*

From this description, Dr. Byrn testifies that "the atropine eye drops discussed by *Chia* teach 'an ophthalmic composition.'" Ex. 1002 ¶ 85. Dr. Byrn further testifies that "a POSA would have recognized that *Chia*'s 'prepackaged' composition, administered over the course of a multi-year study, would almost certainly have included components to reduce degradation of atropine—i.e., a POSA would have recognized that the formulation in *Chia* was stabilized." *Id.* ¶ 86 (citing Ex. 1003, 2). We agree with Dr. Byrn's assessment of the teachings of Chia and credit his testimony.[14]

Petitioner relies on Chia for teaching limitation 1[b]: "from about 0.001 wt % to about 0.05 wt % of atropine or atropine sulfate and water." Pet. 29–30. Petitioner relies on the statement in Chia that "[a]tropine 0.01 % has minimal side effects compared with atropine at 0.1 % and 0.5%, and retains comparable efficacy in controlling myopia progression," as teaching an aqueous 0.01 % atropine solution that "falls squarely within the claimed range of limitation 1[b]. *Id.* at 29–30; Ex. 1002 ¶ 89.

We agree with Petitioner that the overlap between the concentration of Chia's atropine solution and the claimed range establishes a prima facie case of obviousness, for which Patent Owner has not offered any rebuttal. *See Genentech, Inc. v. Hospira Inc.*, 946 F.3d 1333, 1341 (Fed. Cir. 2020) (citing *In re Peterson*, 315 F.3d 1325, 1329 (Fed. Cir. 2003)).

Concerning limitation 1[d]: "the stabilized ophthalmic composition is a liquid," we agree that both Chia and Akorn teach aqueous, i.e., liquid solutions. *See* Pet. 32–33 (citing Ex. 1002 ¶¶ 101–103; Ex. 1004, 1–5; Ex. 1003, 1). As Petitioner states, "[b]oth *Chia* and *Akorn* disclose topical administration of atropine solutions by instillation of eyedrops, and a POSA

[14] Dr. Byrn also relies on Akorn's disclosure of "an FDA-approved, commercially available atropine sulfate ophthalmic solution," as also teaching a stabilized ophthalmic composition for treating myopia. Ex. 1002 ¶ 87. Dr. Byrn testifies that "[a] POSA would have understood that *Akorn* discloses a stabilized ophthalmic composition via its disclosure of several buffering agents to adjust pH to a range of 3.5–6.0, including monobasic sodium phosphate, hydrochloric acid and/or sodium hydroxide." *Id.* Because whether Akorn discloses buffers or a pH range of 3.5–6.0 for its solution is disputed by Patent Owner, we will address Akorn's teachings when we address Patent Owner's criticisms of Petitioner's case below.

would have understood that each solution would be a liquid." Pet. 33 (citing Ex. 1002 ¶¶ 102–103; Ex. 1019, 49). We agree with Dr. Byrn that a POSA would have understood that the ophthalmic composition in Chia and Akorn were liquids.

Limitation 1[e] requires that "the stabilized ophthalmic composition comprises less than about 10% of a degradant formed from degradation of the atropine or atropine sulfate after an extended period of time of at least 2 weeks under a storage temperature of from about 20° C. to about 70° C. and relative humidity from about 50% to about 80%." Ex. 1001, 97:38–44. Petitioner asserts that limitation 1[e] is "merely an obvious, known, and/or inherent property of the obvious formulation claims in elements 1[a]–[d]." Pet. 34. Petitioner further states that the percentage of "a degradant," such as tropic acid from the breakdown of atropine in an aqueous solution, "is simply a property inherent to the atropine or atropine sulfate ophthalmic solution." Pet. 33 (citing Ex. 1002 ¶¶ 43, 105). Petitioner relies on the teachings of Kondritzer to show that limitation 1[d] is an inherent property of the composition, i.e., that the limit on the amount of degradation products under the claimed conditions necessarily results from the claimed stabilized ophthalmic composition. *See* Pet. 34; *see In re Kao*, 639 F.3d 1057, 1070 (Fed. Cir. 2011) (determining claim limitation directed to an inherent property of a formulation "adds nothing of patentable consequence").

For instance, Dr. Byrn testifies that "*Kondritzer* shows that atropine degradation was a predictable function of pH and temperature. Ex. 1005, 1, 4–5. *Kondritzer* derives an equation to predict the pH at which atropine would be most stable for any hydrogen ion concentration. Ex. 1005, 2." Ex. 1002 ¶ 106. Dr. Byrn also testifies that Kondritzer shows in Figure 10 that "at 20° C and pH 5, the half-life of atropine is 266 years; and at 20° C and pH 6, the half-life of atropine is 27 years." Ex. 1002 ¶ 107. Dr. Byrn utilized Kondritzer's chart in Figure 10 to calculate the time it would take for 10% of atropine to degrade at a pH of 5 and 20° C. *Id.* Dr. Byrn concludes that "at 20° C and pH 5, a POSA would reasonably expect the compositions of claim 1 to comprise 90% atropine for approximately 40.4 years. At these parameters, which fall within the limitations of claim 1, *Kondritzer* shows that atropine would readily exceed the stability requirements of limitation 1[e], which allows for up to 10% degradation after at least two weeks." *Id.* ¶ 108.

We credit Dr. Byrn's testimony concerning the teachings of Kondritzer and the ability of a POSA to predict the amount of degradation of atropine over time.[15] Therefore, we find that Petitioner has shown that limitation 1[e] is inherent in the stabilized ophthalmic solution of claim 1.

(2) Limitation 1[c]

The heart of the parties' dispute in this case involves whether the cited art teaches limitation 1[c] that requires "the stabilized ophthalmic

[15] Dr. Laskar faults Dr. Byrn's statement that the humidity of the ambient air "would not be expected to affect the stability of atropine molecules that are in solution." Ex. 2003 ¶ 131; *see* PO Resp. 55–56. Dr. Laskar states that "Dr. Byrn should be aware that the relative humidity of the ambient air can impact the rate of evaporation of the water in the atropine solution, which could then impact the concentration of atropine," especially when stored in a plastic dropper bottle that is semi-permeable to water. Ex. 2003 ¶ 131. Dr. Byrn's testimony, which we credit, finds that the expected degradation of an atropine solution of claim 1 would not only retain 90% atropine for the required two weeks, but for 40.4 years. It is hard to imagine in what scenario the relative humidity could so diminish the stability of an atropine solution made in accordance with claim 1 that it would fail to meet the 2-week limitation to preserve at least 90% of the atropine.

composition further comprises a buffering agent to provide a pH of from about 4.8 to about 6.4." *See* Pet. 30–32; PO Resp. 26–43.

Petitioner relies on Akorn, asserting it discloses "a stabilized ophthalmic composition" that further discloses "several buffering agents to 'provide a pH of from about 4.8 to 6.4,' as in claim 1." Pet. 30–31.

Petitioner further asserts that:

> *Kondritzer* would have provided a reasonable expectation of success to use *Akorn*'s pH range of 3.5–6.0. Ex. 1002 ¶¶ 96–99. *Kondritzer* discloses that optimum atropine stability in aqueous solutions is at pH 3 to 5, and predicts atropine stability in the pH range of 2 to 7. Ex. 1005, 4; Ex. 1002 ¶ 96. However, overly acidic pH was known to be a problem for patient comfort in ophthalmic solutions. Ex. 1002 ¶ 97. A POSA would therefore have targeted the higher end of *Akorn*'s stability range and been motivated to use a pH range of about 5 to about 6 to optimized stability in view of patient comfort. Ex. 1002 ¶¶ 98–99; Ex. 1019, 52 (ophthalmic preparations should be formulated at a pH equivalent to the tear fluid value of 7.4); *see also* Ex. 1005, 4 (showing decreasing atropine stability as pH increases to 7.0). Thus, a POSA would have recognized that a pH range of approximately 3.5–6.0 was optimal. Ex. 1002 ¶ 99.

Id. 31. Petitioner concludes that "[b]ecause pH was a well-known result-effective variable, and because these ranges significantly overlap with the claimed pH range of 4.8 to about 6.4, the claimed range would have been obvious." *Id*. at 32 (citations omitted).

Patent Owner disagrees. Patent Owner asserts that "all grounds critically hinge on using the same 'buffer system' and pH range allegedly disclosed in [Akorn]." PO Resp. 30. But, Patent Owner asserts that Akorn does not state that it employs a buffering agent to control or maintain pH of its atropine solution, and does not even disclose the actual pH of its solution. *Id*. at 26. Patent Owner also asserts that the phosphate salts disclosed in

Akorn are not identified as buffering agents, and could not have buffered Akorn's solution at the listed pH range. *Id*. at 27. Dr. Laskar testifies that in listing the pH range of 3.5 to 6.0, Akorn is merely showing compliance with the pH limits imposed by the National Formulary of the U.S. Pharmacopeia ("USP") and does not state the pH of the described solution. PO Resp. 1, 33–35. Patent Owner also asserts that there is no indication in Akorn that the listed phosphate salts act as buffers, so these could not have "provided" the pH range as required by the claims. *Id*. at 34–35.

Akorn describes the ingredients of its formulation as follows.

> Each mL of Atropine Sulfate Ophthalmic Solution USP, 1% contains: **Active**: atropine sulfate 10 mg equivalent to 8.3 mg of atropine. **Inactives:** benzalkonium chloride 0.1 mg (0.01%), dibasic sodium phosphate, edetate disodium Hypromellose (2910), monobasic sodium phosphate, hydrochloric acid and/or sodium hydroxide may be added to adjust pH (3.5 to 6.0), and water for injection USP.

Ex. 1004, 3.

Patent Owner's argument rests on a listed buffering range in the Waterman reference for phosphoric acid that does not encompass a pH of 3.5 to 6.0 as set forth in Akorn. *See* Ex. 2003 ¶ 52; Ex. 1040, 23. As Dr. Laskar testifies, "Waterman Table 8 lists the 'Buffering Range' of Phosphoric acid as '2–3.1, 6.2–8.2.'" Ex. 2003 ¶ 52 (citing Ex. 1040, Table 8). Dr. Laskar testifies:

> As the Waterman reference submitted with the petition explains, when selecting a buffer, which acts to maintain a solution pH (e.g., at the pH of maximum stability of the active ingredient), the "primary criteria for selection" are "the pH range and buffer capacity [which is maximal at a pH value equal to the pK_a of the buffer (164)] for the buffer." Ex. 1040, 23 (bracketed in material in original). Waterman states that its "Table 8 lists commonly used buffers along with their pK_a

> values and optimal pH ranges." *Id*. Waterman Table 8 lists the "Buffering Range" of Phosphoric acid as "2–3.1, 6.2–8.2." Id., Table 8; *see also* paragraph 9 above. In other words, the buffering range of phosphoric acid and its dibasic sodium phosphate and monobasic sodium phosphate salts does not cover or even overlap with a solution pH of 3.5–6.0.

Ex. 2003 ¶ 52.

Patent Owner's argument presumes that the pK_a for a particular buffer is set at a particular pH range which is not malleable depending on the conditions of the solution that is being buffered. Waterman itself refutes this presumption. First, Waterman describes the buffering ranges listed in Table 8 as "optimal" pH ranges of $pK_a \pm 1$ indicating some buffering capacity may exist below or above the given ranges. *See* Ex. 1040, 23. Dr. Byrn testifies that the most effective pH range for a buffer is within ± 1 of its pK_a, but "[i]t's not a hard cutoff; it's an effectiveness issue." Ex. 2009, 87:5–12; *see id*. at 88:23–89:19 ($pK_a \pm 1$ covers the most effective buffer range, but it is not a hard cutoff), 91:7–18 (person of skill knows $pK_a \pm 1$ is not a hard cutoff); 95:11–97:4 (applying the Henderson, Hasselbach equation provides a wider pH buffering range than $pK_a \pm 1$). Dr. Laskar testifies that pK_a changes as a function of temperature. Ex. 1078, 124:5–9. Waterman also states that "[u]se of co-solvents, surfactants, and complexing agents to solubilize a drug may also influence that buffering capacity and final pH of a formulation by altering the effective pK_a of the buffer and/or directly interacting with the buffer components." Ex. 1040, 24. The evidence of record indicates that the pK_a for a buffer is not static, but depends on the parameters of the solution in which the buffer is used.

As for the specific phosphate buffers used in Akorn and claim 7 of the '787 patent, Petitioner submitted credible evidence that the lower end of the buffer range for these buffers may be as low as pH 5.4. For instance, in Chapter 4 of the Fifth Edition of the European Pharmacopoeia, with which Dr. Laskar testifies he is familiar and which contains recipes for various buffer solutions, is described a 0.067 molar phosphate buffer solution with a pH of 5.4 that is within the 3.5 to 6.0 pH range set forth in Akorn. *See* Ex. 1081, 6; Ex. 1078, 113:12–118:20; *see also* Ex. 1078, 115:22–116:5 (stating European Pharmacopoeia provides a recipe for an 0.067 molar phosphate buffer solution pH 5.4 within the 3.5 to 6.0 pH range).

Petitioner also presents further evidence of a phosphate buffer with a pH range of 5.8 to 8.0 as set forth in a book titled "Buffers: A guide for the preparation and use of buffer in biological systems," by Calbiochem, an affiliate of Merck KGaA. *See* Ex. 1082, 1, 2, 25. The excerpt showing the recipe is set forth in a screenshot, which is reproduced below. Ex. 1082, 25.

6. Phosphate Buffer; pH range 5.8 to 8.0

(a) 0.1 M Sodium phosphate monobasic; 13.8 g/l (monohydrate, M.W. 138.0)
(b) 0.1 M Sodium phosphate dibasic; 26.8 g/l (heptahydrate, M.W. 268.0)

Mix sodium phosphate monobasic and dibasic solutions in the proportions indicated and adjust the final volume to 200 ml with deionized water. Adjust the final pH using a sensitive pH meter.

ml of Sodium phosphate, Monobasic	92.0	81.5	73.5	62.5	51.0	39.0	28.0	19.0	13.0	8.5	5.3
ml of Sodium phosphate, Dibasic	8.0	18.5	26.5	37.5	49.0	61.0	72.0	81.0	87.0	91.5	94.7
pH	5.8	6.2	6.4	6.6	6.8	7.0	7.2	7.4	7.6	7.8	8.0

The screenshot above shows a recipe and accompanying table for making an aqueous phosphate buffer having a pH between 5.8 and 8.0 by mixing 0.1 M sodium phosphate monobasic with 0.1 M sodium phosphate dibasic in deionized water.

Petitioner further presents evidence from PubChem from the National Institutes of Health on the pK_a values for Monosodium phosphate and disodium hydrogen phosphate. *See* Exs. 1084, 1085. The first publication lists the pK_a for monosodium phosphate as a range from 6.8–7.2 "depending on the physiochemical characteristics during pK_a determination." Ex. 1084, 7. The second publication lists the pK_a's for disodium hydrogen phosphate in the following table, which differ from those set forth in the Waterman reference. Ex. 1085, 8.

3.2.13 Dissociation Constants

pKa1 = 2.15; pKa2 = 6.83; pKa3 = 12.38 (phosphoric acid), all at 25 °C

Sigma-Aldrich; Sodium phosphate dibasic. S9763. Sigma-Aldrich, St. Louis, MO. Available from, as of Nov 30, 2006: https://www.sigmaaldrich.com/catalog/search/ProductDetail/SIAL/S9763

The table is table 3.2.13 from the PubChem listing for Disodium hydrogen phosphate, showing pK_a values for dissociation of 2.15, 6.83, and 12.38 at 25° C.

With this record evidence in mind, we turn to what a POSA would glean from reviewing Akorn. Dr. Byrn testifies that Akorn discloses standard buffering agents, including dibasic sodium phosphate, also known as disodium hydrogen phosphate, and monobasic sodium phosphate, also known as sodium dihydrogen phosphate, that have been shown in the evidence just discussed to be able to buffer Akorn's formulation at least in the upper part of the cited pH range. Ex. 1002 ¶ 94–95 (citing Ex. 1019, 52; Ex. 1026, 4–5). We disagree with Dr. Laskar's reliance solely on the Waterman reference as disclosing a buffering range for phosphate salts that "has no overlap with a solution pH of 3.5 to 6.0," and do not credit his conclusion that a POSA would understand this lack of overlap "to completely contradict Dr. Byrn's assertions that these phosphate salts represent a 'common buffer' that was used in EX1004 'to maintain a pH of 3.5 to 6.0.'" *See* Ex. 2003 ¶ 78.

We do not agree with Dr. Laskar's criticism that Akorn does not disclose buffers. *See* Ex. 2003 ¶¶ 71–72. Dr. Laskar testifies that Akorn does not disclose any buffers. *Id*. Dr. Laskar further testifies:

> To the contrary, the disclosure of EX1004 is consistent with the direction provided in Remington and the state of the art in 2015 regarding atropine aqueous ophthalmic solutions. Everything I have read in EX1004 is consistent with the state of the art before 2015, which was to avoid using a buffer or buffering agent in the acidic solution of storage within the boundaries (pH 3.5–6.0) imposed by the USP.

Ex. 2003 ¶ 71.

We find, however, that Dr. Laskar's statements concerning the state of the art in 2015, including about the teachings of the Remington reference, are not sufficiently supported by record evidence. For instance, Dr. Laskar premises his conclusion concerning the state of the art in 2015 that buffers should not be used for atropine solutions, as informed by the Remington reference Dr. Laskar calls the formulator's "bible," on the mistaken belief that the phosphate buffers listed in Akorn cannot buffer below a pH of 6.0. *See generally* Ex. 2003 ¶¶ 45–69. Remington, however, expressly lists phosphate salts in a suggested ophthalmic solution for atropine, but Dr. Laskar states, without support other than the Waterman reference, that "[n]either one of these phosphate salts, alone or in the disclosed combination, is a buffering agent to provide a pH of from about 4.8 to about 6.4. Nor do they provide, alone or in combination, a buffer for a solution having a pH no lower than 3.5 and no higher than 6.0." *Id*. ¶ 50.

As set forth above, we do not agree that the phosphate salts listed in Remington do not buffer at a pH of 6.0 or below. We do not understand Dr. Laskar's statement that these buffers also do not provide a pH from about 4.8 to about 6.4. Dr. Laskar testifies that it is difficult to change the pH of a solution by adjusting the ratio of acid to conjugate base of a buffer from a 50:50 ratio, *see* Ex. 2003 ¶ 59, and states that "[i]n Remington, for example, the atropine ophthalmic vehicle uses somewhat more than twice as much monobasic phosphate as dibasic phosphate on a molar basis, but this alters the pH only by about 0.3 pH units," *id*. Dr. Laskar concludes, however, that the phosphate buffer system listed in claim 7 of the '787 patent, the same buffering pair listed in Akorn, *can* provide a pH no greater than about 6.4 by utilizing "extremely large and counterintuitive variations from standard buffer protocols" to be able to "buffer near the limits of an agent's buffering range." *Id*. Patent Owner cannot have it both ways; phosphate buffers listed in Remington in the suggested atropine solution cannot be said to not buffer, while phosphate buffers listed in claim 7 of the '787 patent, which do not have any particular amount or ratio of acid to base listed, do perform the function of a buffer that provides a pH of from about 4.8 to about 6.4.

We agree with Dr. Byrn that Remington does recommend the use of buffers to maintain the pH at the appropriate level. Ex. 1002 ¶ 50 (citing Ex. 1019, 52–54). In discussing ophthalmic preparations, Remington states:

> Ideally, ophthalmic preparations should be formulated at a pH equivalent to the tear fluid value of 7.4. Practically, this seldom is achieved. The large majority of active ingredients used in ophthalmology are salts of weak bases and are most stable at an acid pH. . . .
>
> Optimum pH adjustment generally requires a compromise on the part of the formulator. The pH selected should be optimum for stability. The buffer system selected should have a capacity adequate to maintain pH within the stability range for the duration of the product shelf life. Buffer capacity is the key in this situation.

> It generally is accepted that a low (acid) pH *per se* necessarily will not cause stinging or discomfort on instillation. If the overall pH of the tears, after instillation, reverts rapidly to pH 7.4, discomfort is minimal. On the other hand, if the buffer capacity is sufficient to resist adjustment by tear fluid and the overall eye pH remains acid for an appreciable period of time, then stinging and discomfort may result. Consequently, buffer capacity should be adequate for stability, but minimized so far as possible, to allow the overall pH of the tear fluid to be disrupted only momentarily.

Ex. 1019, 54.

Remington is recommending a buffer to stabilize an ophthalmic solution for the product's shelf life, but also recommends a balance with the buffers capacity to prevent any resistance to adjustment by tear fluid when the ophthalmic solution is applied to the eye to provide patient comfort and compliance. Contrary to Dr. Laskar's testimony, the record evidence shows that buffers are recommended for ophthalmic solutions to maintain shelf life.

Akorn also discloses a pH range of 3.5 to 6.0 for its formulation that overlaps with the pH range at which atropine was known to be most stable, pH 3–5. Ex. 1002 ¶ 93; *see* Ex. 2003 ¶ 45 (Dr. Laskar admits that "The United States Pharmacopeia/National Formulary had long required atropine sulfate ophthalmic solution to be packaged for storage at a solution pH no greater than 6.0 and no less than 3.5."). Patent Owner takes issue with the fact that we don't know the exact pH of Akorn's formulation. *See* Ex. 2003 ¶ 74. We don't find this argument persuasive as Akorn teaches a particular pH range for its formulation for 1% atropine, which informs a POSA of the appropriate pH range, if not a particular point within the range. The appropriate pH range set forth in Akorn overlaps with the claimed pH range in claim 1, and there is no dispute that pH is a result effective variable for the stability of an atropine solution and finding the optimum pH value for an atropine solution is within the level of ordinary skill of the art. *See* Ex. 1002 ¶¶ 96–98; Ex. 1005 (showing how to predict half-lives of atropine solutions at various pH values and temperatures); Ex. 2003 ¶¶ 38, 61; Ex. 1078, 143:11–144:5 (stating "a POSA would be able to prepare a buffer at pHs within that range of 4.8 to 6.4). This evidence is sufficient to support a case of obviousness. *See E.I. DuPont de Nemours & Co. v. Synvina C.V.*, 904 F.3d 996, 1006 (Fed. Cir. 2018) ("'[W]here there is a range disclosed in the prior art, and the claimed invention falls within that range, the burden of production falls upon the patentee to come forward with evidence'" of teaching away, unexpected results, or other pertinent evidence of nonobviousness.").

Patent Owner also alleges criticality of the claimed pH range set forth in the claims when providing an overview of the '787 patent. *See* PO Resp. 15–19; *see* Ex. 2003 ¶¶ 156–158. Patent Owner argues that the inventors discovered that "using a buffer was critical to atropine stability" for the claimed low dose in the claimed range. PO Resp. 18. This is not appropriate objective evidence of nonobviousness to rebut Petitioner's proof of overlapping ranges between the prior art and the claimed pH range because regardless of whether the problem of the stability of atropine at low doses was unexpected, the result of using a buffer, pH stability for the range, was not unexpected. The buffer is just doing exactly what a buffer does as recognized: *viz.*, maintaining pH. *See* Ex. 1038, 2 (stating "primary purpose of a buffer is to control the pH of the solution"). The fact that the claimed range for a stable low dose atropine solution substantially overlapped with the USP stated range for atropine solutions is also not surprising or unexpected.

(3) Reason to Combine with a Reasonable Expectation of Success

The Petitioner asserts that a POSA would have been motivated to formulate lower-concentration formulations such as the claimed composition in claim 1 to reduce "the side effects without significant loss of efficacy." Pet. 24 (citing Ex. 1002 ¶¶ 67–68). Recognizing the side effects of higher concentration atropine solutions, such as photophobia, cycloplegia, and mydriasis, causing poor patient compliance, Petitioner asserts that Chia studied lower dose atropine compositions and concluded that "a nightly dose of atropine at 0.01% seems to be a safe and effective regimen for slowing myopia progression in children, with minimal impact on visual function in children." *Id*. at 25 (citing Ex. 1002 ¶ 69; Ex. 1003, 1, 7, 8). This conclusion, Petitioner asserts would have motivated a POSA "to pursue a stable, ready-to-use formulation for that treatment." *Id*. (citing Ex. 1003, 8; Ex. 1002 ¶¶ 71–72).

Petitioner further reasons that because using atropine to treat myopia requires long-term treatment, a POSA "would have been motivated to increase the stability and shelf life of low-concentration formulations." Pet. 25 (citing Ex 1003, 1; Ex. 1002 ¶¶ 71–72). In light of the teachings of Kondritzer, Petitioner states that a POSA would have known that atropine's stability was pH dependent, and "[i]n selecting a suitable buffer system for long-term stability of atropine solutions, a POSA would have looked to FDA-approved atropine ophthalmic solutions," such as Akorn. Pet. 25–26 (citing Ex. 1005; Ex. 1002 ¶¶ 43–45, 74). In addition to shelf life, Petitioner asserts a POSA also would have been concerned about patient comfort and compliance with using the low-concentration atropine product and would know that the pH for optimum patient comfort is the same pH as tear fluid, about 7.4. *Id*. at 26–27. In balancing stability of atropine, which is most stable at a pH from 3 to 5 according to Kondritzer, Petitioner asserts that "a POSA would have been motivated to increase the long-term stability of *Chia*'s low-concentration atropine at pHs closer to the clinically desirable pH of 7.4." *Id*. at 27 (citing Ex. 1002 ¶ 78). Petitioner asserts that a POSA would have had a reasonable expectation of success in doing so by using buffers, a common component in ophthalmic compositions, to target "the higher end of Akorn's disclosed pH range in order to optimize stability in view of patient comfort . . . ." *Id*. at 27–28.

Patent Owner responds that the low-concentration atropine formulations did not contain buffers and that the state of the art in 2015 taught excluding the use of buffers. *See* PO Resp. 43–46. We have addressed Patent Owner's argument that the state of the art in 2015 for ophthalmic solutions did not teach buffers. As we found above, the record evidence such as Remington shows that buffers are recommended for ophthalmic solutions to maintain shelf life.

We also note that the prior art references such as Chia described studies to determine the safety and efficacy of low-dose atropine solutions and were not commercially available formulations that must meet FDA requirements for safety and efficacy. Chia specifically states that "[a]tropine 0.01% is currently not commercially available." Ex. 1003, 8.

Patent Owner also asserts that adding a buffer would decrease patient comfort by maintaining a low pH of the atropine solution when administered, causing increased tearing which would undermine efficacy by washing away the atropine. PO Resp. 46–48. We do not find this argument persuasive. As Remington explained, a POSA must choose a buffer to stabilize an ophthalmic solution, but also balance the shelf-life of the solution with a buffering capacity that allows a patient's tears to easily overcome an uncomfortable acidic pH of the ophthalmic solution. *See* Ex. 1019, 54. As Remington states, "[b]uffer capacity is the key," and "buffer capacity should be adequate for stability, but minimized so far as possible to allow the overall pH of the tear fluid to be disrupted only momentarily." *Id*. Buffering does not impede patient comfort when buffer capacity is properly assessed.

We agree with Petitioner and Dr. Byrn that "a POSA would have been motivated to formulate low-concentration atropine solutions that could be administered at pHs closer to those that provided optimal patient compliance and clinical efficacy, but that could also remain stable over long periods of time." Ex. 1002 ¶ 78. We determine that Petitioner has shown an appropriate reason to combine the teachings of Chia, Akorn, and Kondritzer to arrive at the composition of claim 1 with a reasonable expectation of success.

(4) Conclusion

We determine that Petitioner has shown by a preponderance of the evidence that claim 1 of the '787 patent would have been obvious over Chia, Akorn, and Kondritzer.

d) Dependent Claims 2, 4–9, and 12–19

Dependent claims 2, 4, and 5 each recite stabilized ophthalmic compositions with narrow ranges of atropine or atropine sulfate concentrations, i.e., claim 2—"from about 0.001 wt % to about 0.03 wt %," claim 4—"from about 0.01 wt % to about 0.02 wt %," claim 5—"about 0.01 wt %." Petitioner relies on Chia's teaching of 0.01% atropine, which is within claim 2's range, at the lower end of claim 4's range, and is the exact amount as claimed in claim 5, to establish that these dependent claims would have been obvious as well. *See* Pet. 35–36 (citing Ex. 1002 ¶¶ 110–113).

Patent Owner does not address these additional limitations apart from its arguments as to claim 1. The reasons that we set forth above for the determination that claim 1 would have been obvious equally apply here. We determine that Petitioner has shown by the preponderance of the evidence on the record before us that claims 2, 4, and 5 would have been obvious over Chia, Akorn, and Kondritzer.

Claims 6 and 7 further define the buffering agent of claim 1 to include a phosphate buffering agent for claim 6 and "sodium dihydrogen phosphate, disodium hydrogen phosphate, or a combination thereof" for claim 7, which Petitioner asserts Akorn teaches. *See* Pet. 36 (citing Ex. 1002 ¶¶ 116–117; Ex. 1004, 3). Patent Owner reasserts the same argument that it raised with respect to claim 1, namely, that Akorn does not teach the claimed phosphate buffers. *See* PO Resp. 57. We have addressed Patent Owner's arguments with respect to claim 1 and found that Akorn does teach the claimed phosphate buffers.

Claim 8 further requires a tonicity adjusting agent, and claim 9 further defines the tonicity agent as a halide salt of a monovalent cation. Ex. 1001, 98:11–16. Petitioner asserts that "[a] POSA would have recognized that decreasing the concentration of atropine sulfate from *Akorn*'s 1% atropine sulfate solution to *Chia*'s 0.01% atropine sulfate solution would require the addition of a tonicity adjusting agent to maintain eye comfort" due to a reduction in the amount of salt in the solution. Pet. 37 (citing Ex. 1002 ¶¶ 122–124). Petitioner also asserts that sodium chloride, a halide salt of a monovalent cation, is a very common tonicity adjusting agent. *Id.* (citing Ex. 1002 ¶¶ 123–124; Ex. 1036, 1–3; Ex. 1019, 54).

Patent Owner responds that Petitioner has not shown that a reduction from 1% to 0.01% atropine sulfate will make enough difference in the osmolarity to be material to the tonicity of the solution, and Petitioner failed to account for the existing tonicity of Akorn's formulation. PO Resp. 56.

Dr. Byrn explains tonicity and that "isotonicity always is desirable and particularly is important in intraocular solutions." Ex. 1002 ¶ 122 (citing Ex. 1019, 54). Dr. Byrn then testifies that a reduction in the concentration of atropine sulfate, a salt, from 1% to 0.01 % would require a tonicity adjusting agent for eye comfort. *Id.* We credit Dr. Byrn's testimony as it is reasonable that a 100-fold decrease in the concentration of atropine sulfate in the solution would require some tonicity adjustment.

Patent Owner does not address the additional limitations of claims 12–18 apart from its arguments as to claim 1. Claims 12, 13, and 14 further require, respectively, a preservative, in a particular concentration, and a list of particular preservatives, which Chia and/or Akorn teach. *See* Pet. 38–39 (citing Ex. 1002 ¶¶ 126–132; Ex. 1003, 2; Ex. 1022 1–2; Ex. 1023, 1). Claim 15 further requires that the stabilized ophthalmic composition is "essentially free of procaine and benactyzine, or pharmaceutically acceptable salts thereof," which Petitioner asserts is taught by the absence of these ingredients in Chia and Akorn. *See* Pet. 39 (citing Ex. 1002 ¶¶ 133–138; Ex. 1003; Ex. 1004, 3; Ex. 1019, 52). Claims 16 and 17 further require topical administration and instillation, respectively, which Petitioner asserts is taught by Chia and Akorn. *See* Pet. 40 (citing Ex. 1002 ¶¶ 139–143; Ex. 1003, 1; Ex. 1004, 1–4). Claim 18 further requires administering the atropine composition through an eye drop bottle, which Petitioner asserts is disclosed by Akorn. Pet. 41 (citing Ex. 1002 ¶¶ 144–148; Ex. 1004, 1–4).

We have reviewed Petitioner's assertions in the Petition concerning claims 12–18 and Dr. Byrn's testimony in support, and determine that Petitioner has shown by a preponderance of the evidence that these claims would have been obvious over Chia, Akorn, and Kondritzer.

Dependent claim 19 recites: "A method of treating the pre-myopia, myopia, or progression of myopia in an individual in need thereof, comprising administering to an eye of the individual an effective amount of the stabilized ophthalmic composition of claim 1." Ex. 1001, 98:52–57. Petitioner asserts that claim 19's requirement for a method of treating is taught by Chia which found "that administration of its 0.01% atropine solution, via eyedrops, was effective to treat myopia and progression of myopia." Pet. 41 (quoting Ex. 1003, 1).

Patent Owner raises similar arguments as it did for claim 1 that Chia was an unbuffered solution and that buffering decreased patient comfort. *See* Pet. 56–57. We have addressed these arguments with regard to claim 1 and find them unavailing here as well.

We determine that Petitioner has shown by a preponderance of the evidence that claim 19 would have been obvious over Chia, Akorn, and Kondritzer.

E. Remaining Grounds

Because we have determined that all challenged claims are unpatentable as obvious over Chia, Akorn, and Kondritzer, we need not reach the issue of whether these same claims are also unpatentable under Grounds 2 and 3. Therefore, we do not reach these grounds.

III. CONCLUSION[16]

For the foregoing reasons, we conclude Petitioner has shown by a preponderance of the evidence that claims 1, 2, 4–9, and 12–19 of the '787 patent are unpatentable.

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Patent Owner's Motion to Exclude Evidence is *denied*;

FURTHER ORDERED that claims 1, 2, 4–9, 12–19 of U.S. Patent No. 10,842,787 B2 have been shown by a preponderance of the evidence to be unpatentable under 35 U.S.C. § 103; and

FURTHER ORDERED that because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

In summary:

[16] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding. See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1, 2, 4–9, 12–19	103	Chia, Akorn, Kondritzer	1, 2, 4–9, 12–19	
1, 2, 4–9, 12–19	103	Chia, Akorn, Lund[17]		
1, 2, 4–9, 12–19	103	Akorn, Wu, Kondritzer[18]		
Overall Outcome			1, 2, 4–9, 12–19	

[17] Because we find these claims are unpatentable over Chia, Akorn, and Kondritzer, we do not determine their patentability under this ground.
[18] Because we find these claims are unpatentable over Chia, Akorn, and Kondritzer, we do not determine their patentability under this ground.

FOR PETITIONER:

Peter Cuomo
Williams Dixon
Thomas Wintner
MINTZ, LEVIN, COHN, FERRIS, GLOVSKY & POPEO, P.C.
pjcuomo@mintz.com
wsdixon@mintz.com
twintner@mintz.com

FOR PATENT OWNER:

Michael Rosato
Lora Green
Jad Mills
Michael Hostetler
WILSON SONSINI GOODRICH & ROSATI
mrosato@wsgr.com
lgreen@wsgr.com
jmills@wsgr.com
mhostetler@wsgr.com